

### Appropriateness of Endpoints

1. Weight at 6 months: The first outcome measure is best used when a simple comparison of final weights is sufficient for the study. For example, if baseline weights are very similar across the study groups or if interest is on absolute weight achieved at 6 months. It treats the follow-up weight as the primary endpoint (Voils et al., 2017).
2. Change in weight (kg) from baseline to 6 months: This outcome is intuitive for a study on weight-loss. Outcome 2 directly quantifies how many kilograms each participant lost or gained during the study. It is best used when we want to emphasize the change due to the intervention (Jebb et al., 2011).
3. Change in weight from baseline to 6 months, adjusted for baseline weights: Outcome 3 combines outcome 2, where we wished to quantify change in weight, with a statistical adjustment for baseline weight. It is appropriate when we suspect baseline weight has an influence on the amount of weight change, and we want to control for that influence in the analysis (Van Breukelen, 2006).
4. Weight at 6 months, adjusted for baseline weight: This is a common for trials with baseline measurements, where we analyze the primary outcome at the end of the study with baseline as a covariate. This outcome is especially useful in trials where the aim is to describe the final weights of patients in the study, rather than the change in weight, whilst still accounting for different participant weights at study entry .
5. Percentage change in weight from baseline: This outcome is a relative measure of the participants weight loss to their initial body weight. It is often considered when differences in body size are important, such as when a 5 kg loss might be very significant for a 60 kg person but less so for a 150 kg person (Vickers, 2001).

### Advantages and Disadvantages of Primary Endpoints

| Outcome                                                                   | Advantages                                                                                                                                                             | Disadvantages                                                                                                                                                                                               |
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| Weight (kg) at 6 months                                                   | Simple study design with easy to obtain measurements and straightforward analysis of the data.(Voils et al., 2017)                                                     | Does not take into account different baseline weights of participants. Results are less clinically intuitive since there is no measure of how much weight was lost.(Voils et al., 2017)                     |
| Change in weight (kg) from baseline to 6 months                           | Results based on this outcome are easily interpretable. This method accounts for within-person variability, which would reduce bias in the results.(Jebb et al., 2011) | If the intervention and control groups differ slightly in starting weight by chance, analysing weight difference without further adjustment might still be biased or less precise.(Clifton & Clifton, 2019) |
| Change in weight from baseline to 6 months, adjusted for baseline weights | Any random baseline imbalance is adjusted for in the model, improving comparability. Including                                                                         | If baseline weight is very strongly correlated with weight change (or if all groups have very similar baselines), the inclusion of                                                                          |

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|                                                  | baseline weight as a covariate ensures the comparison of weight loss between groups is not skewed by initial weights, which in turn yield unbiased results even if these baseline weights differ (Van Breukelen, 2006).                      | baseline in the model may not dramatically improve precision compared to simpler models (Vickers, 2001).                                                                                                                                                                                                                              |
| Weight at 6 months, adjusted for baseline weight | ANCOVA with baseline has the greatest efficiency. Any random differences in starting weight between groups are accounted for, yielding an “apples-to-apples” comparison (Van Breukelen, 2006).                                               | ANCOVA assumes the relationship between baseline weight and final weight is linear and consistent across groups (no baseline-by-treatment interaction). If lighter or heavier individuals respond differently to the intervention, one might need to include interaction terms or consider stratified analysis (Van Breukelen, 2006). |
| Percentage change in weight from baseline        | By scaling weight change to baseline weight, it puts participants on a more comparable scale. Percent weight change is independent of absolute body size, which might implicitly account for differences in build or height (Vickers, 2001). | Analysing percentage change is statistically less efficient than all the other options. Percent changes often have a skewed distribution. If one group by chance starts heavier, percent change could either exaggerate or underestimate the true effect (Vickers, 2001).                                                             |

## Statistical Analysis

| Outcome                                                                   | Statistical Analysis                                                                                                                                                                                                                                                                                                                              |
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| Weight (kg) at 6 months                                                   | Compare mean weight at 6 month between the intervention and control group using an independent t-test or, equivalently, a linear model with treatment group as a predictor. Weights should be approximately normally distributed. Deviations from normality can be dealt with using transformations if needed (Earnest, 2016).                    |
| Change in weight (kg) from baseline to 6 months                           | Calculate weight change as (weight at 6 months)-(weight at baseline). A two-sample t-test or an equivalent linear model is then used to assess if any change in weight is statistically significant. If data are shown to not be normally distributed, then non-parametric test should be considered or variable transformations (Earnest, 2016). |
| Change in weight from baseline to 6 months, adjusted for baseline weights | Analysis of Covariance (ANCOVA) is used with weight change as the outcome, baseline weight as a covariate, and treatment group as the main independent variable. One would check the assumption that baseline weight has a linear association with weight change (Van Breukelen, 2006).                                                           |
| Weight at 6 months, adjusted for baseline weight                          | Conduct an Analysis of Covariance (ANCOVA) with 6-month weight as the dependent variable, treatment group as the independent factor, and baseline weight as a covariate. We will verify no violation of model assumptions, such as linearity, homoscedasticity, normality of residuals (Van Breukelen, 2006).                                     |
| Percentage change in weight from baseline                                 | Each participant's % weight loss = (Weight change in kg / Baseline weight) × 100 is computed. The primary analysis would compare the mean percentage loss between groups using an independent t-test or ANOVA or a non-parametric test if the distribution is highly skewed (Vickers, 2001).                                                      |

## Sample Size Considerations

We assume a two-arm trial (1:1 allocation), two-sided  $\alpha=0.05$ , 80% power. A clinically meaningful weight loss is often taken as about 3–5% of baseline weight (roughly 3–5 kg for a 100 kg adult) (Foley et al., 2016). We use  $\Delta=5$  kg for weight outcomes and  $\Delta=5\%$  for percentage weight change. Typical SDs are large for body weight (10–12 kg) but smaller for change scores (6.5 kg for 6-month weight change); percentage changes have SD of order 6–7%. Baseline and 6-month weight are positively correlated in obesity trials (e.g.  $\rho\approx 0.55$ ). We will illustrate calculations using  $\sigma\approx 12$  kg (final weight),  $\sigma\approx 6.5$  kg (weight change), and  $\rho\approx 0.6$  ( $R^2\approx 0.36$ ) for covariate-adjusted analyses.

For a continuous outcome comparing two independent means with equal group sizes (options 1, 2, and 5), the required sample size per group, as suggested in Chow et al. (2017) is given by the standard formula:

$$n = \frac{2 (Z_{1-\alpha/2} + Z_{1-\beta})^2 \sigma^2}{\Delta^2}$$

where  $\Delta$  is the true difference to detect,  $\sigma^2$  is the (assumed common) variance, and  $Z_{1-\alpha/2} = 1.96$ ,  $Z_{1-\beta} = 0.84$  for 90% power.

For ANCOVA analysis (options 3 and 4) adjusting for baseline, Borm et al. (2007) show that the effective sample size calculated using the above formula is multiplied by  $1 - \rho^2$ , where  $\rho$  is the correlation between baseline and follow-up outcome.

$$n_{\text{ANCOVA}} = (1 - \rho^2) n_{\text{unadj}}$$

## Recommendation

Weight at 6 months, adjusted for baseline weight, is recommended as the primary outcome for this trial. This method provides a balance of statistical efficiency and clarity for a weight loss study. By controlling for participants' starting weights, the variance in outcomes is reduced, allowing the detection of smaller differences with the given sample size. This approach provides an unbiased estimate of the treatment effect, even if there are chance imbalances in baseline weight.

## References

- Borm, G. F., Fransen, J., & Lemmens, W. A. J. G. (2007). A simple sample size formula for analysis of covariance in randomized clinical trials. *Journal of Clinical Epidemiology*, 60(12), 1234–1238. <https://doi.org/10.1016/j.jclinepi.2007.02.006>
- Chow, S.-C., Shao, J., Wang, H., & Lokhnygina, Y. (2017). *Sample size calculations in clinical research*. Chapman and Hall/CRC.
- Clifton, L., & Clifton, D. A. (2019). The correlation between baseline score and post-intervention score, and its implications for statistical analysis. *Trials*, 20(1), 43. <https://doi.org/10.1186/s13063-018-3108-3>
- Earnest, A. (2016). *Essentials of a successful biostatistical collaboration*. Chapman and Hall/CRC.
- Foley, P., Steinberg, D., Levine, E., Askew, S., Batch, B. C., Puleo, E. M., Svetkey, L. P., Bosworth, H. B., DeVries, A., Miranda, H., & Bennett, G. G. (2016). Track: A randomized controlled trial of a digital health obesity treatment intervention for medically vulnerable primary care patients. *Contemporary Clinical Trials*, 48, 12–20. <https://doi.org/10.1016/j.cct.2016.03.006>
- Jebb, S. A., Ahern, A. L., Olson, A. D., Aston, L. M., Holzapfel, C., Stoll, J., Amann-Gassner, U., Simpson, A. E., Fuller, N. R., Pearson, S., Lau, N. S., Mander, A. P., Hauner, H., & Caterson, I. D. (2011). Primary care referral to a commercial provider for weight loss treatment versus standard care: a randomised controlled trial. *The Lancet*, 378(9801), 1485–1492. [https://doi.org/10.1016/S0140-6736\(11\)61344-5](https://doi.org/10.1016/S0140-6736(11)61344-5)
- Van Breukelen, G. J. P. (2006). ANCOVA versus change from baseline: more power in randomized studies, more bias in nonrandomized studies [corrected]. *Journal of Clinical Epidemiology*, 59(9), 920–925. <https://doi.org/10.1016/j.jclinepi.2006.02.007>
- Vickers, A. J. (2001). The use of percentage change from baseline as an outcome in a controlled trial is statistically inefficient: a simulation study. *BMC Medical Research Methodology*, 1(1), 6. <https://doi.org/10.1186/1471-2288-1-6>
- Voils, C. I., Olsen, M. K., Gierisch, J. M., McVay, M. A., Grubber, J. M., Gaillard, L., Bolton, J., Maciejewski, M. L., Strawbridge, E., & Yancy, W. S. J. (2017). Maintenance of Weight Loss After Initiation of Nutrition Training: A Randomized Trial. *Annals of Internal Medicine*, 166(7), 463–471. <https://doi.org/10.7326/M16-2160>